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Selenizing *Hericium erinaceus* polysaccharides induces dendritic cells maturation through MAPK and NF-κB signaling pathways

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ABSTRACT

In this study, polysaccharides extracted from *Hericium erinaceus* were modified to obtain its nine selenium derivatives, $sHEP_1$-$sHEP_9$. Their structures were identified, yields and selenium contents were determined, the phenotypic and functional maturation of murine bone marrow-derived dendritic cells (DCs) and relevant mechanisms were compared taking unmodified HEP as control. The results revealed that the selenylation were successful. $sHEP_1$, $sHEP_2$ and $sHEP_8$ treatment of DCs increased their surface expression of MHC-II and CD86 and indicated that $sHEP_1$, $sHEP_2$ and $sHEP_8$ induced DC maturation. Furthermore, $sHEP_2$ and $sHEP_8$ also significantly decreased DCs endocytosis and significantly enhanced cytokine (IL-12 and IFN-γ) production. In line with TLR4 activation, $sHEP_2$ increased the phosphorylation of ERK, p38, and JNK, and the nuclear translocation of p-c-Jun, p-CREB, and c-Fos. $sHEP_2$ also activated NF-κB signaling, as evidenced by degradation of IκBα/β and nuclear translocation of p65 and p50. Together, these results suggest that sHEP is a strong immunostimulant.

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1. Introduction

Hericium erinaceus has been used as a commom edible and medicinal mushroom in East Asian. It is very popular for traditional cuisines, especially in Japan and China [1]. It is a traditional medicinal food for the prevention and treatment of chronic atrophic gastritis (CAG), duodenal ulcers and several other diseases [2]. Many researchers have demonstrated that *Hericium erinaceus* contains some important pharmacological constituents. The polysaccharides derived from *Hericium erinaceus* have various biological activities such as anti-inflammatory, anti-microbial [3], antioxidant [4,5], antitumor [6] and immunomodulatory [7]. The average molecular weight of HES was 22 kDa composing of rhamnose, mannose, galactose, and glucose in a molar ratio of 1:1.1:16.2:36.8.

In recent years, using HNO_3-Na_2SeO_3 method to modify polysaccharide has proved to be an effective way. This method is simple and production is fast and selenium content of modifier is higher [8]. Many researchers have demonstrated that selenylation modification polysaccharides have many different biological activities such as antioxidant [9], immune-enhancing [10–12], anti-hyperglycemia [13] and so on.

Dendritic cells (DCs) are the most potent antigen-presenting cells that initiate immune responses. Immature DCs originate from lymphoid or myeloid progenitors, which are located at different sites of the body, encountering the invading foreign antigens or pathogens [14]. After maturation, DCs possess very strong antigen presentation through up-regulating the major histocompatibility complex (MHC) and co-stimulating molecule expression, and strongly stimulating T cells [15]. DCs could secrete cytokines like IL-12 to activate NK cell and act as anticancer agents. Besides these increased data showed that DCs were responsible for the induction of immunological tolerance [16,17]. In addition, there are several exogenous inducers of DC maturation, such as Toll-like receptor (TLR)2, TLR4, and TLR9, and so on [18]. In particular, TLR4 is in mammals, which could mediate MyD88 dependent form and MyD88 independent form signaling pathway. And many natural polysaccharides isolated from fungi such as Coriolus versicolor mushroom [19], Ganoderma lucidum [20], Polyporus umbellatus [21], seem to induce DC maturation by mediating TLR4 binding.

In this study, we measured the basic characteristics of sHEP obtained by the purification were estimated using Gel Permeation Chromatography (GPC) and infrared spectroscopy (IR). At the same

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time, we evaluated the in vitro effects of selenizing *Hericium erinaceus* polysaccharides (sHEP) on dendritic cells. In this way, we provide a theoretical basis for the development of a novel polysaccharide immunopotentiator.

2. Materials and methods

2.1. Plant material

Hericium erinaceus (HE) were purchased in November, 2014 from Huishihao Pharmacy in Fujian Province of China. The voucher specimen (No. HP20140910) was deposited in Key Laboratory of Traditional Chinese Veterinary Medicine and Animal Health in Fujian province, Fujian Agriculture and Forestry University.

2.2. Chemicals and reference compounds

Nitric acid (HNO_3) was the product of Shanghai Lingfeng Chemical Reagent Ltd. Sodium selenite bought from Shanghai Lingfeng Chemical Reagent Ltd. Standardry selenium stored solution at 100 $\mu g\,mL^{-1}$ supplied by National standard substance research center. Hydrochloric acid (HCl) was the product of Nanjing Chemical Reagent Ltd. Perchloric acid ($HClO_4$) was the product of Tianjin Xinyuan Chemical Reagent Ltd.

GM-CSF, rmIL-4 and Lipopolysaccharide (LPS) were purchased from Sigma-Aldrich (St. Louis, MO, USA). Anti-mouse antibodies against CD11c, CD86, and MHC-II were purchased from BD Pharmingen (San Diego, CA, USA) and those against extracellular signal-regulated kinase (ERK), C-Jun N-terminal kinase (JNK), and p38 mitogen-activated protein kinases (MAPKs) were purchased from Cell Signaling Technology (Beverly, MA, USA).

2.3. Preparation of H. erinaceus polysaccharides

Dried HE (1000 g) was crushed into 0.3–1 cm^3 small block, soaked 12 h with 2000 mL of 95% ethanol, reflowed for 1 h twice in water bath of 80 °C. After aired 12 h, the drug was decocted with 20-fold volume water 3 times each for 30 min. The physic liquor was filtrated through two-layers gauze, concentrated into 1000 mL, centrifugated at 2500 rpm for 20 min and added with 95% ethanol up to 90% of concentration (v/v). After maintaining 24 h, the precipitation was dried by vacuum freeze-drying machine (Model LGJ-25, Dongxing MachineryIndustry Co., Ltd. Shamen City). And then the precipitation was lyophilized to obtain the crude HEP.

The crude HEP was eliminated protein by Sevage method [22] and dissolved into 0.05 $g\,mL^{-1}$ with distilled water, added into a chromatographic column of Sephadex G-100 (1.5 cm × 200 cm) and eluted with distilled water. The flow rate was maintained at 12 $mL\,h^{-1}$, the eluent was collected by automatic fraction collector, 4 mL per tube, and measured for polysaccharide by the phenol-sulfuric acid method. The elution curve was drawn (one peak). The eluents contained polysaccharides were merged and lyophilized to get one purified HEP. Its carbohydrate content was 81.3% determined by the phenol-sulfuric acid method [23,24].

2.4. Selenylation modification of HEP

2.4.1. Design of modification condition

Three factors respectively at three levels, the usage amount of sodium selenite at 200, 300 and 400 mg for 500 mg of HEP (A), the reaction temperature at 50, 70 and 90 °C (B) and the reaction time for 6, 8 and 10 h (C), were selected [25]. Nine modification conditions were designed according to orthogonal test as $L_9(3^4)$ (Table 1).

2.4.2. Selenylation reaction

The HNO_3-Na_2SeO_3 method was applied [8]. 4.5 g of HEP was divide equally into 9 portions and added respectively into the three-necked flask filled with 50 mL of 5% HNO_3 solution stirring to make HEP completely dissolve, Then the sodium selenite solution was added and stirring reaction was performed at definitive temperature and duration designed in Table 1. After the reaction finished, the mixture was cooled to room temperature, adjusted pH to 5–6 with saturated sodium carbonate solution, dialyzed in dialysis sack with 1 kDa ultrafiltration membrane against tap water and sampled for determination of sodium selenium every 6 h by ascorbic acid method [25].

2.5. Identification of sHEP

2.5.1. Assay of selenium content

The atomic fluorescence spectrometry was used for determination of selenium content [26] by atomic fluorescence spectrometer (Model AFS-930, Beijing Jitian nstrument Co., Ltd.). The working conditions of the spectrometer were as follows: the negative high voltage was 270 V, lamp current was 80 mA, the atomization temperature was 200 °C, the height of atomization gas was 8 mm, the discharges of carrier gas and shield gas flow were 400 $mL\,min^{-1}$ and 800 $mL\,min^{-1}$ respectively, the injection volume was 1 mL, the readings was peak area, reading time was 7 s, the delay time was 1.5 s. 100 mL of standardry selenium solution and 5% HCl solution as diluent were linked to the atomic fluorescence spectrometer. The concentrations of standard curve were set at 0, 4, 8, 12, 16 and 20 $ng\,mL^{-1}$ which were automatically diluted and the fluorescence intensity were detected by the spectrometer. The standard curve was drawn by taking the selenium mass concentration as abscissa and fluorescence intensity as vertical axis. 20 mg of sHEP weighed accurately was dissolved in 10 mL of ultrapure water, 0.5 mL of sHEP solution was accurately measured and added into triangular flask with cork, 10 mL of $HClO_4$-HNO_3 (v/v, 1:1) mixed acid solution was added to digest for 12 h at 4 °C, then to heat at 100 °C replenishing the mixed acid solution timely. When sHEP solution become clear, colorless and accompanied by white smoke, it was concentrated to about 2 mL, cooled to room temperature, and diluted accurately into 25 mL with 5% HCl solution in which 1 mL of the solution was accurately measured and diluted into 100 mL with 5% HCl solution as sample solution. The same method was used for preparing the blank sample solution. The fluorescence intensities of the sample solution were detected by the spectrometer. The selenium contents were calculated according to the standard curve.

2.5.2. Infrared spectroscopy analysis

The FT-IR spectra of HEP and sHEP in a wavenumber range of 4000–400 cm^{-1} were recorded by KBr pellets method with a Nicolet 200 Magna-IR spectrometer (Nicolet Instrument Corp).

2.5.3. Molecular weight analysis

The average M_W was measured through GPC on an ultrahydrogel linear column at 25 °C. The eluent (flow rate: 1.0 $mL\,min^{-1}$) was a 0.1 M sodium acetate solution. Pullulan standards were used as the standards for M_W measurement (M_W: 6.1–70.8 kDa).

2.5.4. Monosaccharide composition

The monosaccharide composition of the polysaccharides was determined by gas chromatography as described previously [27]. Fructose, rhamnose, arabinose, xylose, glucose, mannose and galactose were used as the monosaccharide standards. The average molecular weight was determined by high-performance lipid chromatography (HPLC) [28].

The monosaccharide composition of the polysaccharides was determined by gas chromatography. The polysaccharides were

Table 1
Modification conditions, yields and contents of selenium and carbohydrate of sHEP.

sHEPS	A Temperature (°C)	B Na_2SeO_3 mg)	C Time (h)	Rate of yeild%	Se content mg g^{-1}	Carbohydrate content(%)
$sHEP_1$	50	200	6	63.71	10.92	78.06
$sHEP_2$	50	300	8	61.11	10.64	66.23
$sHEP_3$	50	400	10	66.64	6.03	77.95
$sHEP_4$	70	200	8	68.00	6.48	64.71
$sHEP_5$	70	300	10	68.05	13.50	74.24
$sHEP_6$	70	400	6	71.75	10.66	55.18
$sHEP_7$	90	200	10	56.62	5.56	79.68
$sHEP_8$	90	300	6	63.32	9.41	77.05
$sHEP_9$	90	400	8	65.25	13.98	80.32

hydrolyzed with 2 M trifluoroacetic acid (120 °C, 3 h). After hydrolysis, the neutral monosaccharides were successively reduced with Sodium borohydride (NaBH4) and acetylated with 1:1 pyridine acetic anhydride at 90 °C for 1 h. Alditole acetates were analyzed by GC on a Shimadzu GC-14C instrument equipped with an Rtx 2330 column (Shimadzu Co. LTD., Kyoto, Japan, 30 m × 0.32 mm × 0.2 m).

2.6. Analysis of a range of sHEP doses on DCs stimulating proliferation of T cells

2.6.1. Generation of immature DCs

ICR mouse (4–6 weeks old) were sacrificed and the femurs and tibias were isolated from the surrounding muscle tissue. The bones were washed twice in PBS (pH 7.4) after the muscle tissue was removed. Clusters within the marrow suspension were collected and centrifuged at 1500 rpm for 20 min at 25 °C to obtain the precipitate. DCs were cultured at a starting concentration of 1.0×10^6 mL^{-1} in round-bottomed 12-well plates with RPMI-1640 supplemented with GM-CSF, rmIL-4, 10% FCS, 2 mL per well. Cells were cultured in a humidified chamber at 37 °C and 5% CO_2. After incubation for 24 h, the medium with non-adherent cells was replaced with fresh medium. The culture medium was removed and replenished with fresh medium every two days. On the 7th day, the matured DCs were harvested for stimulation of following assays.

2.6.2. Allogeneic mixed lymphocyte reaction

On the 7th day of incubation, different concentrations (0.781–12.5 μg mL^{-1}) of sHEP, HEP, RPMI-1640 (serum-free) and LPS (5 μg mL^{-1}) were added into DCs incubated for 48 h. After 30 min, cells were treatment with mitomycin C (50 μg mL^{-1}) at 37 °C, and then washed with PBS twice. The resulting pellet was re-suspended into 5.0×10^5 mL^{-1} with RPMI-1640 media. Spleen from ICR mouse were harvested sterility mashed with the flat surface of a syringe plunger against a stainless steel sieve (200 mesh). After the splenocytes were washed thrice and resuspended in complete RPMI-1640. Splenocytes (1.0×10^6 mL^{-1}) with a volume of 100 μL $well^{-1}$ were cultured in 96-well plates. The mature DCs were added into each well (four wells each group).

2.6.3. Analysis of T cell proliferation

After incubating for 68 h, 20 μL of MTT (5 mg mL^{-1}) was added into each well, and continued to incubate for 4 h. Then supernatant were removed before adding DMSO (100 μL $well^{-1}$). Finally, A_{570} was tested as the index of BMDCs stimulating the proliferation of T cells.

2.7. Ability of antigen presentation assessment of sHEP on DCs

OVA solution mixed with aluminium adjuvant, each mouse was immunized subcutaneous with 100 μg OVA on the 7th, 14th day. Three days after the last immunization, the lymphocytes were collected as in Section 2.6.2. Lymphocytes were resuspended in complete RPMI-1640 at a concentration of 1.0×10^6 mL^{-1}, supplemented with OVA (100 μg mL^{-1}), in 96-well plates. Then DCs having been treated with mitomycin C were added into lymphocytes (four wells each group). After 68 h of the incubation period, 20 μL of MTT (5 mg mL^{-1}) was added into each well, and continued to incubate for 4 h. Then supernatant were removed before DMSO (100 μL $well^{-1}$) were added. Finally, A_{570} was tested as the index of ability of antigen presentation of DCs.

2.8. Measurement of IL-12 and IFN-γ production by ELISA

Various concentrations of sHEP, HEP (0.781–2.5 μg mL^{-1}) were used to stimulate the BMHC-imDCs. After 48 h of treatment, the supernatants were collected by centrifugation at 1000 rpm for 8 min at 4 °C and were stored at −20 °C. An IL-12 and IFN-γ expression was determined using a cytokine assay kit (Calvin Biological Technology Co., Ltd., Suzhou, China) according to the manufacturer's instructions.

2.9. Phenotype analysis

Phenotypic maturation of DCs was analyzed by flow cytometry [29]. Cell staining was performed using a Fluorescein isothiocyanate (FITC)-conjugated anti-body (against CD86 and MHC-II). Cells were analyzed using a BD Accuri flow cytometer, and the data were analyzed using BD Accuri C6 Software. Forward and side scatter parameters were used to gate live cells.

2.10. Endocytosis assay

DCs were collected at the end of the culture with sHEP, HEP or LPS, and then incubated with 1 mg mL^{-1} FITC-dextran (40,000 Da, Sigma) at 37 °C for 1 h. The reaction was stopped by cold PBS containing 2% FBS. The cells were washed three times and then stained with PE-conjugated anti-CD11c antibody for 30 min without light. The double stained DCs were analyzed by BD Accuri flow cytometer.

2.11. Western blotting

Lysates were prepared from total cells or nuclear fractions as previously described [30]. Detergent-insoluble material was removed, and equal amounts of protein were fractionated by 10% sodium dodecyl sulfate-polyacryl-amide gel electrophoresis and transferred to pure nitrocellulose membranes. Membranes were blocked with 5% skim milk in Tris-buffered saline containing 0.05% Tween 20 for 1 h, and then incubated with an appropriate dilution of primary antibodies in 5% BSA (in Tris-buffered saline containing Tween 20) for 2 h. Blots were incubated with biotinylated secondary antibody for 1 h and then with horseradish peroxidase-conjugated streptavidin for 1 h. Signals were detected by enhanced chemiluminescence. Band intensities were quantified using ImageJ 1.47 software.

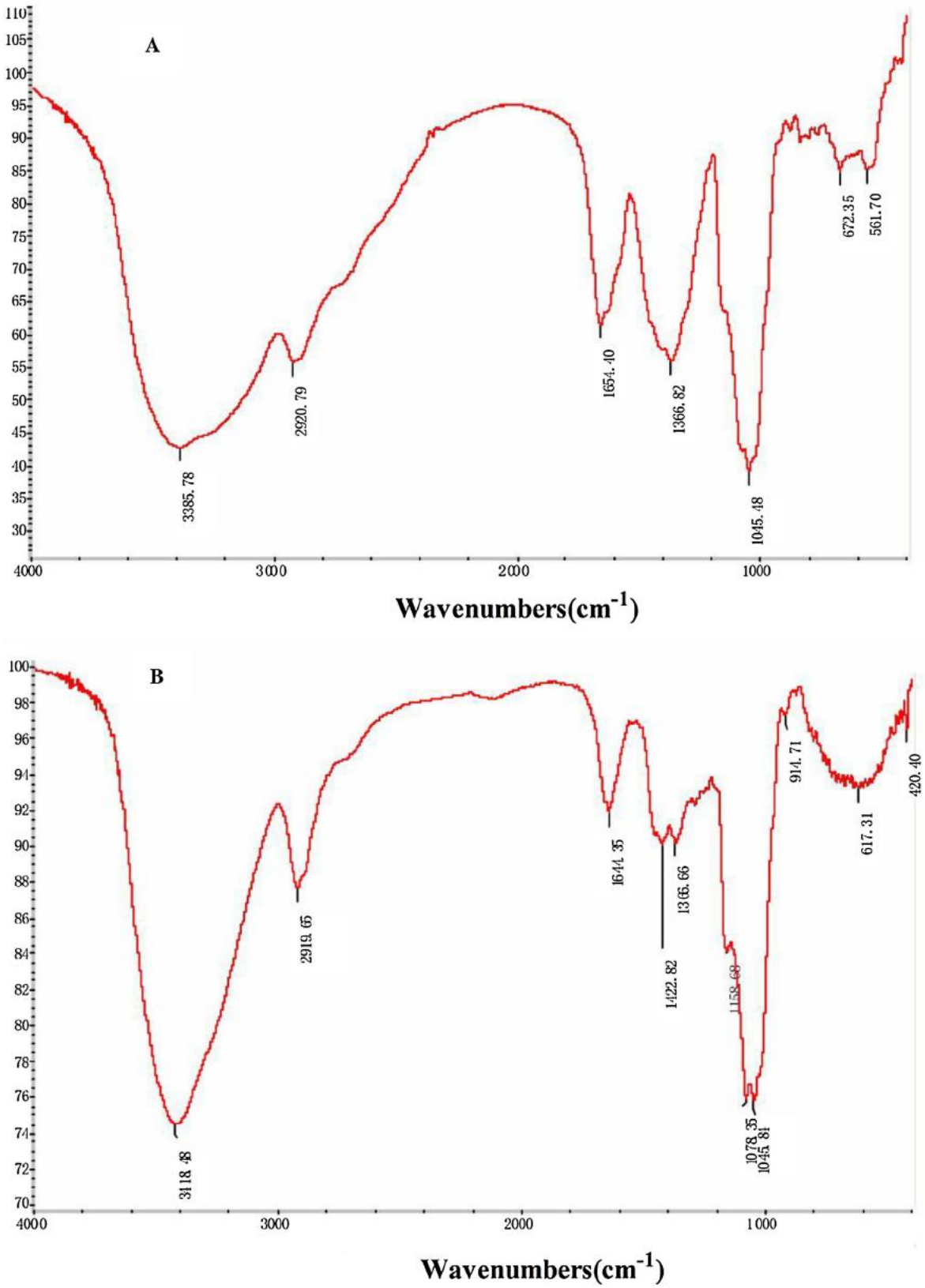

Fig. 1. (A) Infrared spectrum of HEP; (B) Infrared spectrum of sHEP.

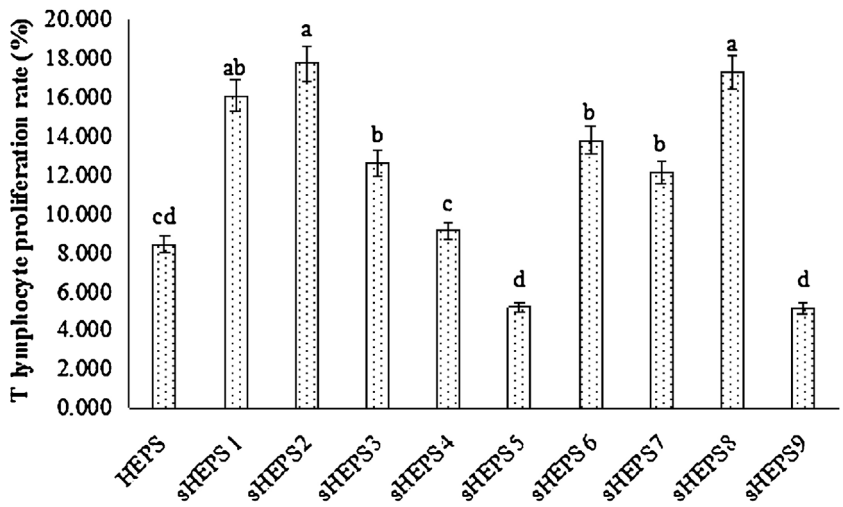

Fig. 2. Effects of HEP and sHEPs on T lymphocyte proliferation rate.

2.12. Statistical analysis

All data obtained in this study were processed statistically and divergence were presented as mean ± SD. SPSS 16.0 for Windows was used and $P < 0.05$ indicated significance differences.

3. Results

3.1. The modification conditions, yields, selenium contents, carbohydrate contents and chemical analysis of sHEPs

The modification conditions, yields, selenium contents and carbohydrate contents of sHEP are listed in Table 1. The yield of $sHEP_6$ was the highest (71.75%), the next were $sHEP_5$, $sHEP_4$ and $sHEP_3$. The selenium content of $sHEP_9$ was the highest (13.98%), next were $sHEP_5$, $sHEP_1$ and $sHEP_6$. The carbohydrate content of $sHEP_9$ was the highest (80.32%), next were $sHEP_7$, $sHEP_1$ and $sHEP_3$.

The M_W was determined through HPLC-GPC. The M_W of HEP and $sHEP_2$ (selected the best) were approximately 16.15 and 22.37 kDa, respectively.

3.2. The infrared spectra of HEP and sHEP

The FT-IR spectra of HEP and sHEP are illustrated in Fig. 1A. In the spectra of HEP and sHEP, there were three band, one appeared in 3600–3200 cm^{-1} corresponding to the hydroxyl stretching vibration, the second appeared in the 2932 cm^{-1} corresponding to methyl C—H stretching vibration, the third appeared in 1400–1000 cm^{-1} corresponding to C—O—C stretching vibration. This indicated that the HEP and sHEP were polysaccharides.

As compared with the spectrum of HEP, the spectrum of sHEP presented two characteristic absorption bands, one appeared at 1078.35 cm^{-1} describing a symmetrical O—Se—O stretching vibration (O—Se—O, 1010–1040 cm^{-1}) and another at 617.31 cm^{-1} describing an asymmetrical Se—O—C stretching vibration (Se—O—C, 600–700 cm^{-1}) (Fig. 1B), which indicated that sHEP was successfully modified in selenylation.

3.3. Monosaccharide composition

The monosaccharide composition of HEP was determined. The result shows that HEP comprises four kinds of monosaccharides: glucose, galactose, mannose and arabinose with a molar ratio of 50.80: 42.30: 4.58: 2.29.

3.4. Effect of sHEP on immunocompetence of DCs

3.4.1. sHEP enhanced the stimulation of DCs on T lymphocytes proliferation rate

The HEP and sHEP enhanced the stimulation of DCs on T lymphocytes proliferation rate are illustrated in Fig. 2. During the single stimulation of polysaccharide, the proliferation rate of $sHEP_2$ group was the highest (17.73%) and the following was $sHEP_8$ group (17.28%) and $sHEP_1$ group (16.09%), these three groups were significantly higher than that of HEP group (8.42%) ($P < 0.05$). Therefore, the $sHEP_1$, $sHEP_2$ and $sHEP_8$ were selected in following experiment.

3.4.2. sHEP improved the antigen presenting ability of DCs

The antigen presenting ability of DCs in each group are listed in Table 2. Compared with LPS group, the A_{570} values was higher between 12.5 and 1.563 $\mu g\,mL^{-1}$ $sHEP_2$ group. The 3.125 and 0.781 $\mu g\,mL^{-1}$ $sHEP_8$ group was significantly higher than that of LPS group ($P < 0.05$). The 1.563 and 0.781 $\mu g\,mL^{-1}$ $sHEP_1$ group was the highest and significantly higher than that of LPS group ($P < 0.05$). Therefore, $sHEP_1$, $sHEP_2$ and sHEP8 at 1.563 and 0.781 $\mu g\,mL^{-1}$ could improve antigen presenting ability of DCs. But the 6.25 $\mu g\,mL^{-1}$ $sHEP_2$ group was the highest and significantly higher than that of other group ($P < 0.05$).

3.4.3. IL-12 and IFN-γ secretion amount changes of DCs

The IL-12 and IFN-γ concentrations of culture supernatant in each group are illustrated in Fig. 3. The IL-12 concentrations in HEP, $sHEP_1$, $sHEP_2$ and $sHEP_8$ at 12.5–1.563 $\mu g\,mL^{-1}$ groups were significantly higher than that in cell control group ($P < 0.05$). The IL-12 concentration in $sHEP_2$ at 6.25 $\mu g\,mL^{-1}$ group was the highest and higher than those in other groups (Fig. 3A). The IFN-γ concentrations in $sHEP_2$ and $sHEP_8$ at 12.5–1.563 $\mu g\,mL^{-1}$ groups were significantly higher than that in cell control group, HEP and $sHEP_1$ groups ($P < 0.05$). The IL-12 concentration in $sHEP_2$ at 6.25 $\mu g\,mL^{-1}$ group was the highest and significantly higher than those in other groups and LPS group ($P < 0.05$) (Fig. 3B).

3.4.4. Effect of HEP and sHEP on DC cell surface molecule expression

The effect of HEP, $sHEP_1$, $sHEP_2$ and $sHEP_8$ (6.25 $\mu g\,mL^{-1}$) of DCs were determined the expression levels of CD86 and MHCII by flow cytometry, and the results were show in Fig. 4. The expression levels of CD86 in $sHEP_2$ group were significantly higher than those in cell control and HEP group ($P < 0.05$). The expression levels of

Table 2
Effects of HEP and sHEP on ability of antigen presenting of DCs (A_{570} values).

Concentration ($\mu g\,mL^{-1}$)	HEP	$sHEP_1$	$sHEP_2$	$sHEP_8$
cc	0.165 ± 0.003^{c}	0.165 ± 0.003^{c}	0.165 ± 0.003^{d}	0.165 ± 0.003^{d}
12.5	0.173 ± 0.001^{b}	0.178 ± 0.003^{b}	0.202 ± 0.018^{b}	0.176 ± 0.005^{bc}
6.25	0.179 ± 0.009^{b}	0.175 ± 0.002^{b}	0.229 ± 0.019^{a}	0.181 ± 0.004^{bc}
3.125	0.175 ± 0.003^{b}	0.175 ± 0.005^{b}	0.209 ± 0.010^{b}	0.209 ± 0.013^{a}
1.563	0.188 ± 0.004^{a}	0.220 ± 0.008^{a}	0.207 ± 0.006^{b}	0.208 ± 0.010^{a}
LPS	0.173 ± 0.002^{b}	0.173 ± 0.002^{b}	0.173 ± 0.002^{c}	0.173 ± 0.002^{c}

Column data marked without the same letters (a-d) differ significantly ($P < 0.05$).

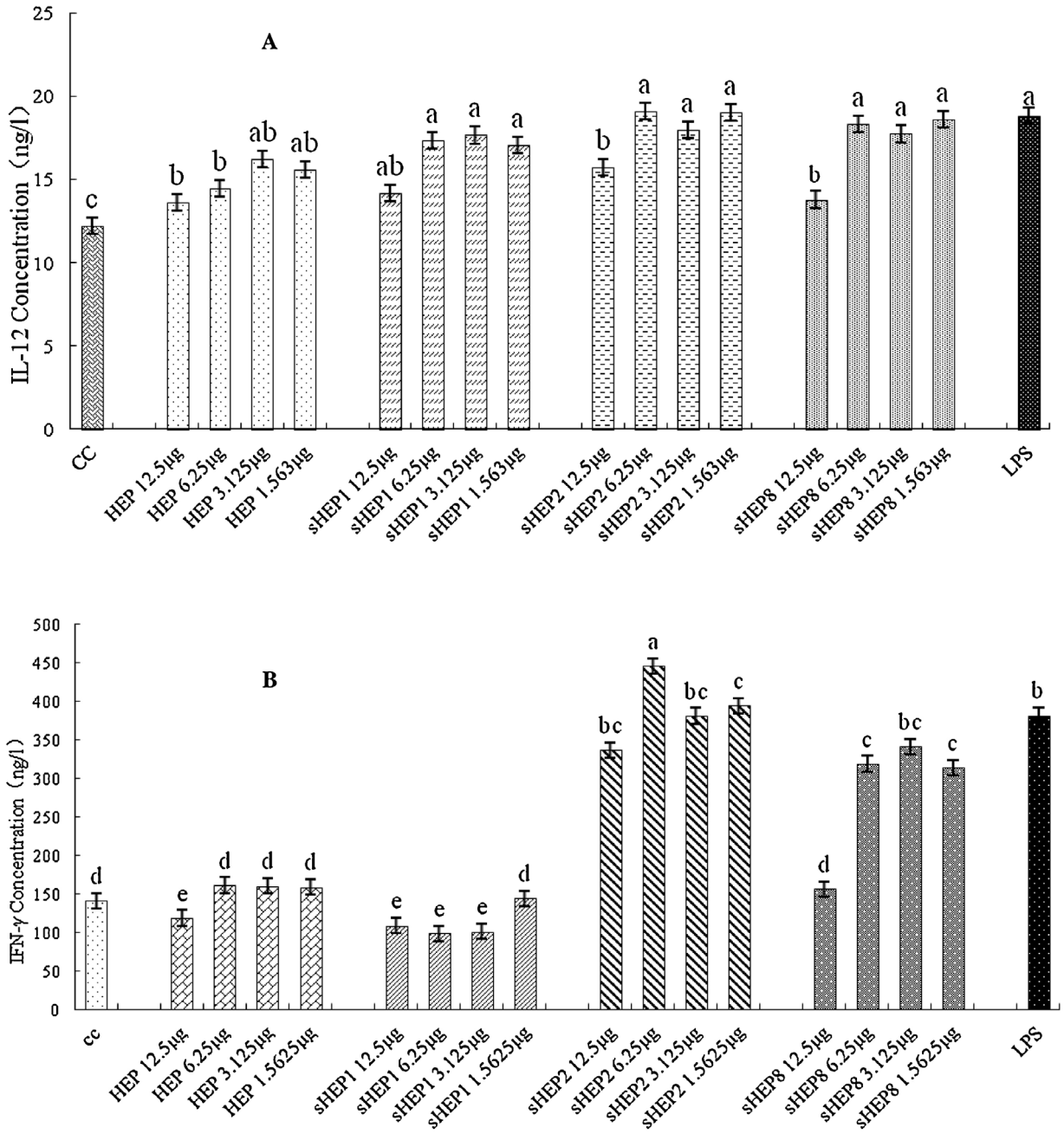

Fig. 3. (A) Effects of HEP and sHEP on IL-12 secretion of DCs; (B) Effects of HEP and sHEP on IFN-γ secretion of DCs.

MHCII in $sHEP_2$ and $sHEP_8$ group were significantly higher than those in cell control and HEP group ($P < 0.05$). It is suggested that the $sHEP_2$ group could enhance the surface molecule activity of the DC cell.

3.4.5. *Effect of HEP and sHEP on endocytosis activity of DCs*

To examined the endocytic activity of DCs treated with HEP and sHEP (6.25 $\mu g\,mL^{-1}$) by monitoring the uptake of FITC-dextran (Fig. 5). The result showed that the endocytic activity in $sHEP_1$,

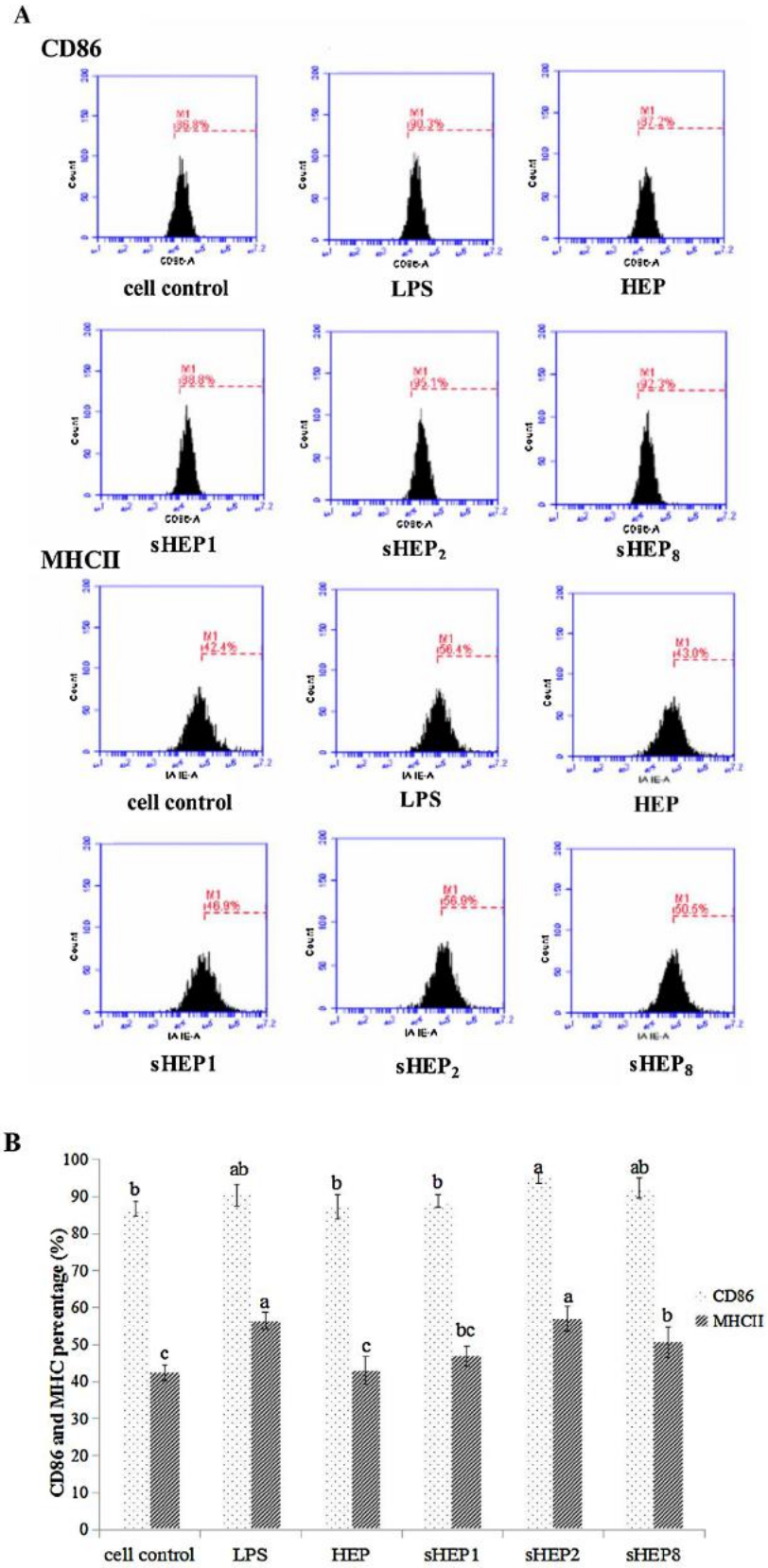

Fig. 4. Effect of HEP and sHEP on cell surface molecule expression on DCs. (A) The expressions of CD86 and MHCII were analyzed by flow cytometry; (B) The expressions of CD86 and MHCII. The value are presented as mean ± SD (n = 4).

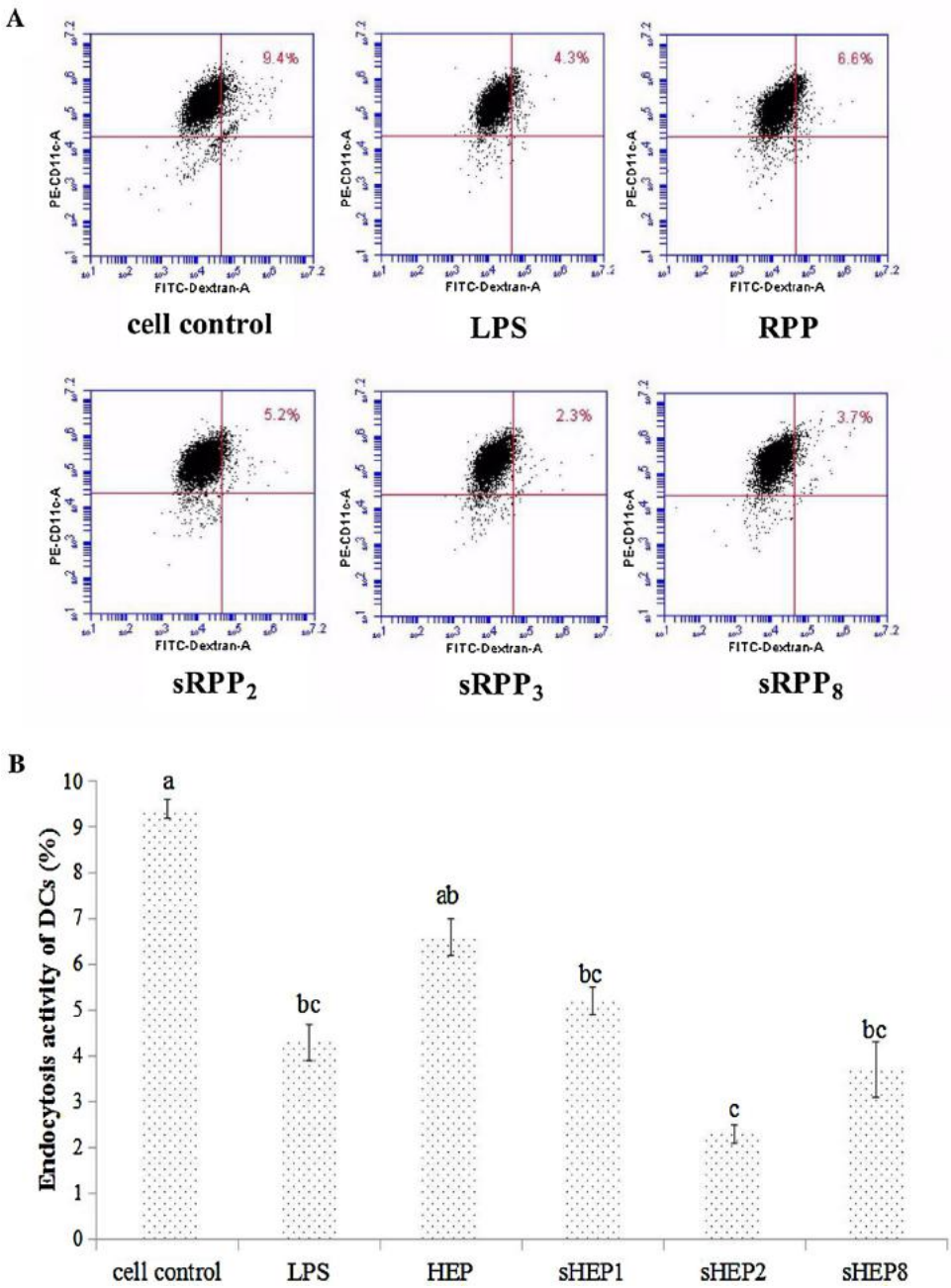

Fig. 5. Effect of HEP and sHEP on endocytosis activity of DCs. (A) The endocytic activity of DCs incubated with FITC-dextran; (B) The endocytic activity of DCs treated with HEP and sHEP. The value are presented as mean ± SD (n = 4).

$sHEP_2$ and $sHEP_8$ group were significantly lower than that in cell control group($P < 0.05$). The endocytic activity in $sHEP_2$ group were significantly lower than that in HEP group ($P < 0.05$).

3.5. Activation of intracellular signaling in HEP/$sHEP_2$-treated DCs

As shown in Fig. 6A, basal levels of phosphorylated p38, ERK, and JNK MAPKs in control group were very low, whereas their phosphorylation was strongly increased by HEP and $sHEP_2$ treatment. Subsequent nuclear translocation of transcription factors p-c-Jun, p-CREB, and c-Fos was also increased in HEP/$sHEP_2$-treated DCs (Fig. 6B). But $sHEP_2$-treated DCs showed an increased phosphorylation more than HEP group.

NF-κB subunits are mostly kept in the cytoplasm by IκBα/β and translocate to the nucleus after IκBα/β degradation. HEP and $sHEP_2$ increased IκBα/β degradation in the cytoplasm (Fig. 6C) and increased nuclear translocation of NF-κB subunits p50 and p65 (Fig. 6D). But $sHEP_2$-treated DCs showed more increased IκBα/β degradation in the cytoplasma and increased NF-κB p50/p65 in the nuclear than HEP group. These data show that HEP and $sHEP_2$ activates MAPK and NF-κB signaling downstream of TLR4. Moreover, the $sHEP_2$ group was stronger than that in HEP group in enhancing the MAPK and NF-κB signaling downstream of TLR4.

4. Discussion

As far as we know, Steinman (Canadian scholar) first identified dentritic cells in 1973. DCs are the predominant professional

antigen-presenting cell, Which play an important role in both inherent immunity and adaptive immunity [31,32]. There are two types of DCs in organism, which are immature and mature DCs. Immature DCs can capture and process antigens. Mature DCs can

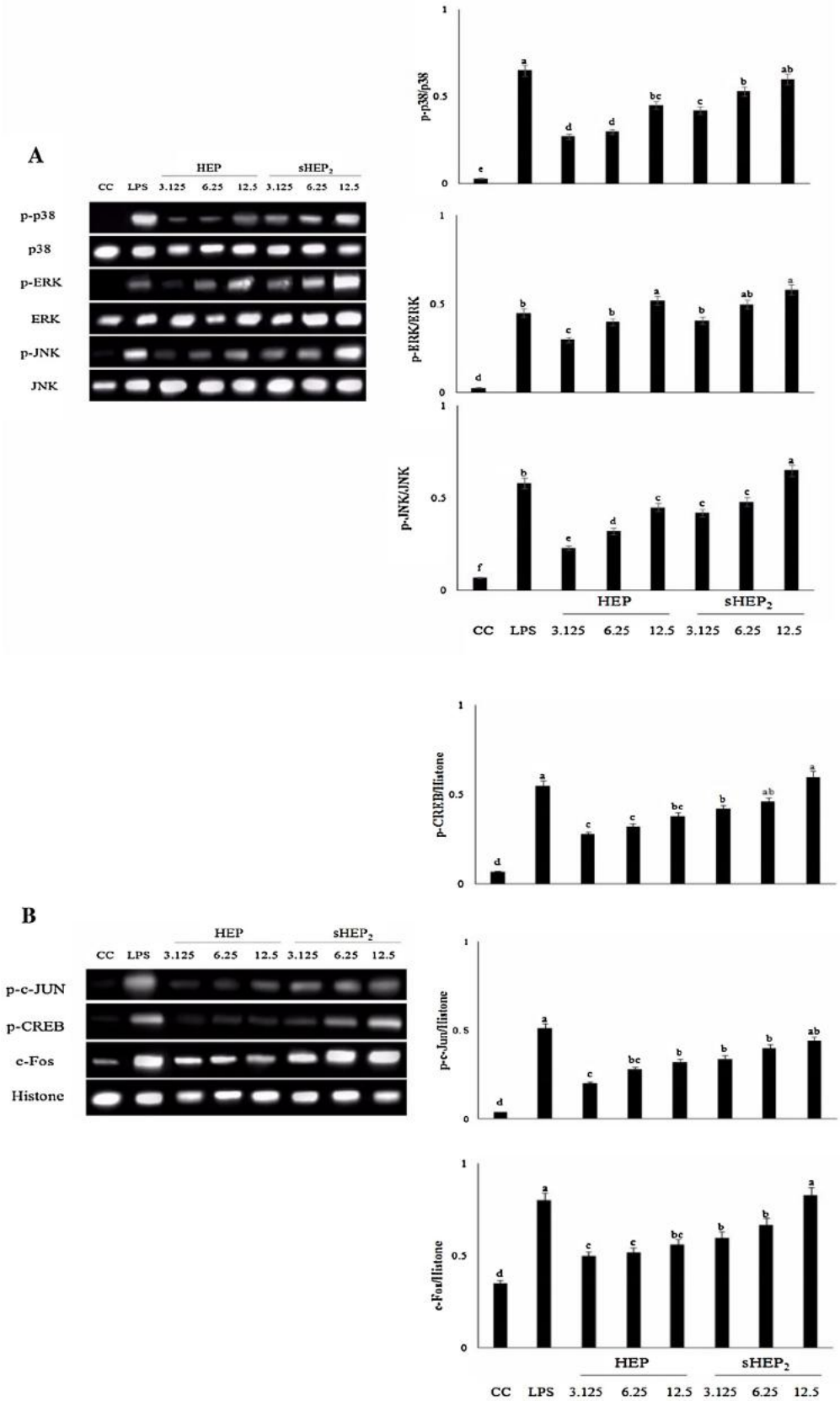

Fig. 6. Effect of HEP and $sHEP_2$ on MAPK and NF-κB signaling in DCs. (A, B) Effect of HEP and $sHEP_2$ on MAPK signaling in DCs, (A) Levels of phosphorylated p38, ERK, and JNK, (B) Levels of p-c-Jun, p-CREB, and c-Fos, (C, D) Effect of HEP and $sHEP_2$ on NF-κB signaling in DCs, (C) Levels of IκBα/β, (B) Levels of p50/p65.

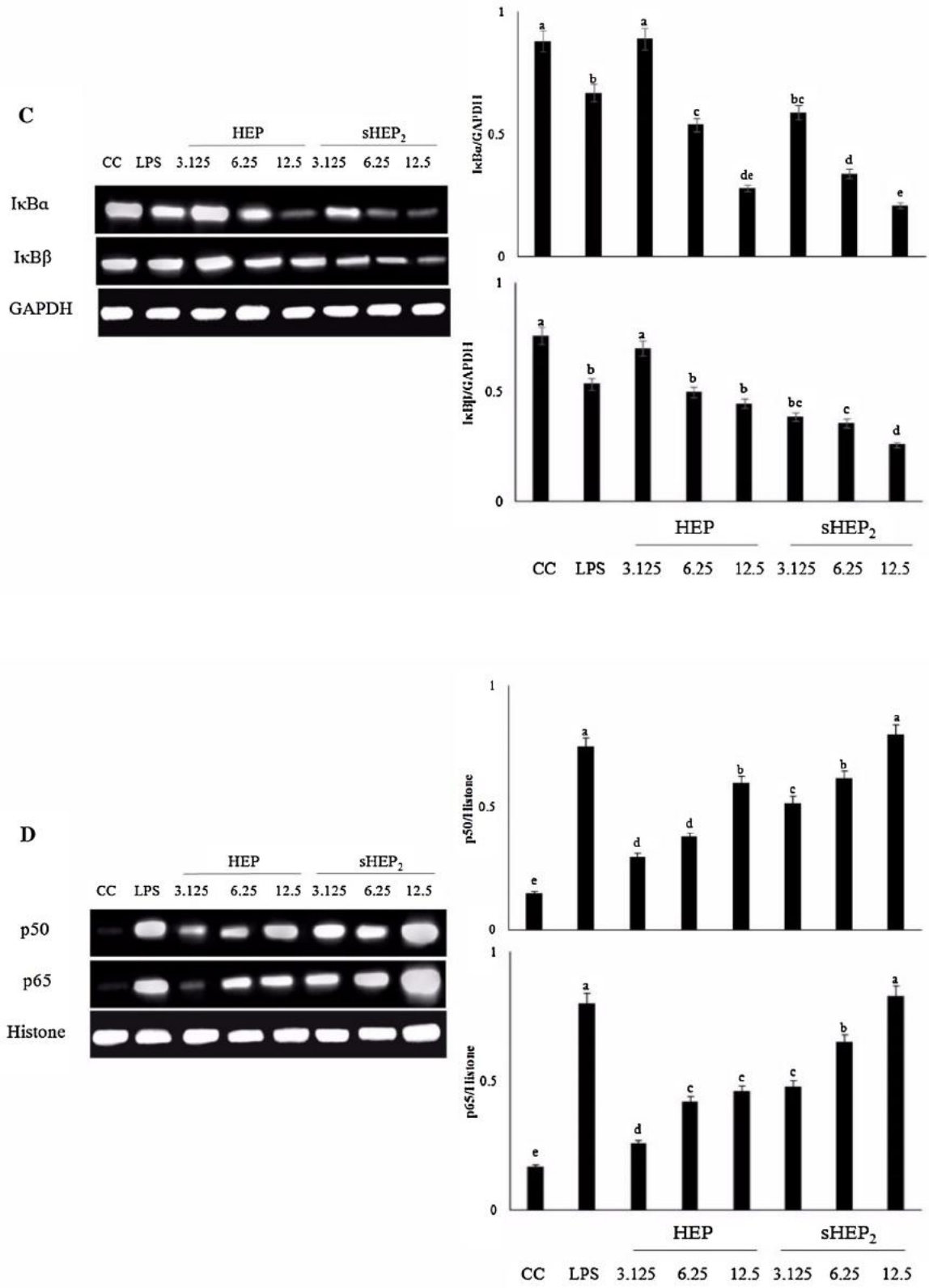

Fig. 6. (*Continued*)

upregulate costimulatory molecules and adhesion molecules. DCs can efficacious activate play a central role in initiating T-cell against pathogens, and play a central role in the adaptive immune response [33,34]. It is greatly significant to study the effect of DCs in the process of inducing, maintaining and regulating immune response. Thus, our research model is DCs in this experiment, which is important for further investigating the immunological function of HEP and sHEP. As shown in Fig. 2, the result showed that $sHEP_1$, $sHEP_2$, $sHEP_8$ could significantly promote the proliferation rate of T lymphocyte compared with HEP when DCs were stimulated with HEP

and sHEP. Furthermore, in this experiment, DCs and T cells both come from ICR mice, thus they have the same MHC-II molecules, which makes testing on its ability of antigen presenting possible. The result indicated that antigen presenting ability of DC had been significantly enhanced after being incubated with $sHEP_1$, $sHEP_2$, $sHEP_8$ (1.563 and 0.781 $\mu g\,mL^{-1}$), and the 6.25 $\mu g\,mL^{-1}$ $sHEP_2$ group was the highest and significantly higher than that of other group (Table 2). These results demonstrated that selenylation modification could significantly enhance the effect of HEP on antigen presenting ability of DC.

A recent study has shown that immature DCs only weakly activate T cells due to relatively low expression of co-stimulatory molecules (CD86) and MHC-II [35]. In the present experiment, the expression levels of CD86 and MHC-II in $sHEP_1$, $sHEP_2$ and $sHEP_8$ group were more intensive than those in cell control and HEP group. Moreover, the $sHEP_2$ group was stronger than that in LPS group in enhancing the surface molecule activity of the DC cell (Fig. 4). These results may be explained sHEP ability to induce the maturation of DCs. Furthermore, it is reported that immature DCs had powerful ability and specialized in antigen capture and processing [7,29], but lose this capacity upon maturation. To examine the effects of HEP and sHEP on endocytosis in DCs, the fluorescent marker FITC-dextran were used. The marker is mainly taken up via the mannose receptor. After incubation of DCs with HEP and sHEP for 24 h, FITC-dextran was added to the culture medium for 1 h. Immature DCs had a high level of endocytosis, while sHEP-treated DCs showed a decreased antigen uptake ability (Fig. 5). These findings suggest that sHEP enhances the phenotypic and functional maturation of DCs.

Cytokines are essential molecules involved in the differentiation, maturation and activation of cells [36]. DCs can also activate T cells via the secretion of cytokines, such as IL-12 and IFN-γ, which promotes the development of the Th1 pathway. IL-12 is known as the most important cytokine for induction of a Th1 response. It induced IFN-γ production in NK and T cells and enhanced the cytolytic activity of these cells [37,38]. IFN-γ is a pleiotropic cytokine with immunomodulatory effects on different kinds of immune cells. In mammals and poultry, IFN-γ has been an indicator for cell-mediated immunity of infected organism [39]. Therefore, preliminary assessment for activation extent of DCs could be made by detecting the content of IFN-γ [40]. Therefore, the concentration changes of IFN-γ, IL-12 were important to immune response. In this research, the result showed that the concentrations of IL-12 and IFN-γ in $sHEP_2$ and $sHEP_8$ were all significantly higher than that of HEP (Fig. 3A). These provided the information that $sHEP_2$ and $sHEP_8$ could promote proliferation and differentiation of Th1 cell, and promote the Th1-medicated cellular immunity. It might also come to the conclusion that $sHEP_2$ and $sHEP_8$ enhancing on Th1-medicated cellular immunity made it the immunoenhancement of adaptive immunity, especially the $sHEP_2$.

NF-κB is kind of transcription factor that plays an important role in DC maturation [41–43]. It could induce the expression of various gene involved in immune responses [44]. The MAPKs signaling pathway also was involved in inducing DCs maturation. It regulates the expression of key surface molecules and the secretion of cytokines [45]. MAPKs and NF-κB signaling pathways regulate phenotypic and functional maturation of DCs. ERK activation regulates DC survival and proliferation, p38 activation induces DC maturation and migration, and NF-κB activation is responsible for DC maturation [46,47]. In this study, basal levels of phosphorylated ERK, p38, and JNK MAPKs in immature DCs were very low, whereas their phosphorylation was strongly increased by HEP and $sHEP_2$ treatment (Fig. 6A). And subsequent nuclear translocation of transcription factors p-c-Jun, p-CREB, and c-Fos was also increased in HEP/$sHEP_2$-treated DCs (Fig. 6B). Moreover, HEP and $sHEP_2$ increased IκBα/β degradation in the cytoplasm (Fig. 6C) and increased nuclear translocation of NF-κB subunits p50 and p65 (Fig. 6D). But these data show that the $sHEP_2$ group was stronger than that in HEP group in enhancing the MAPK and NF-κB signaling pathway. However, many studies report that other transcription factors also involved in DC maturation. Therefore, we will investigate the effect of other transcription in DC maturation in the future.

5. Conclusion

In this study a homogeneous *Hericium erinaceus* polysaccharides were modified to obtain its nine selenium derivatives, $sHEP_1$-$sHEP_9$. This is the first report on the evaluation of the composition and immunoenhancement properties of selenium polysaccharides from *H. erinaceus*. We have shown that sHEP (especially $sHEP_2$) treatment promoted morphological changes of murine DCs and a reduction of DC endocytosis, which are consistent with DC maturation. And sHEP (especially $sHEP_2$) treatment also increased the expression of surface molecules that are important for antigen presentation. Furthermore, sHEP (especially $sHEP_2$) treatment stimulated DCs to secrete cytokines that promote TH1 responses. Finally, sHEP (especially $sHEP_2$) enhance the MAPK and NF-κB signaling downstream of TLR4. Therefore, $sHEP_2$ could be explored as a natural immune stimulant for functional foods.

Conflict of interests

Authors declare that there is no conflict of interests.

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